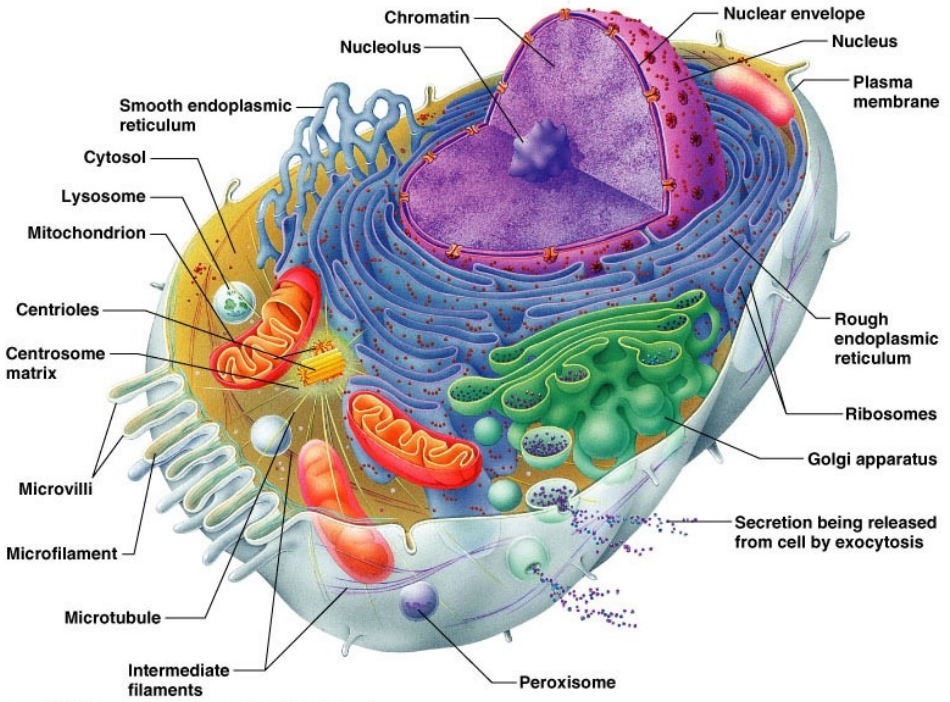
Endomembrane system

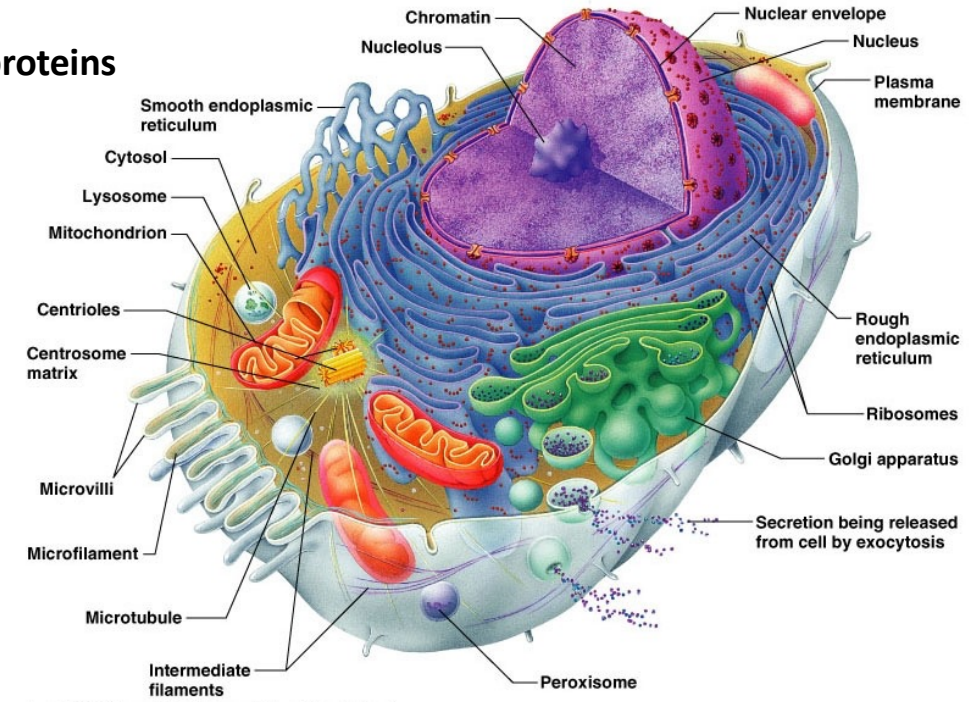
Lecture 3.7 September 10th, 2026



Dr. Serrano | dserrano@unav.es

Endomembrane system

1. Membrane trafficking within cells
2. Endoplasmic reticulum (ER)
    - 2.1 Smooth endoplasmic reticulum (SER)
    - 2.2 Rough endoplasmic reticulum (RER)
    - 2.3 Protein trafficking and destination
    - 2.4 Nuclear translocation of cytosolic proteins
    - 2.5 Protein systhesis in the RER
3. The Golgi complex
    - 3.1 Example: lysosomal enzymes
4. Protein orientation



1 Membrane trafficking within cells

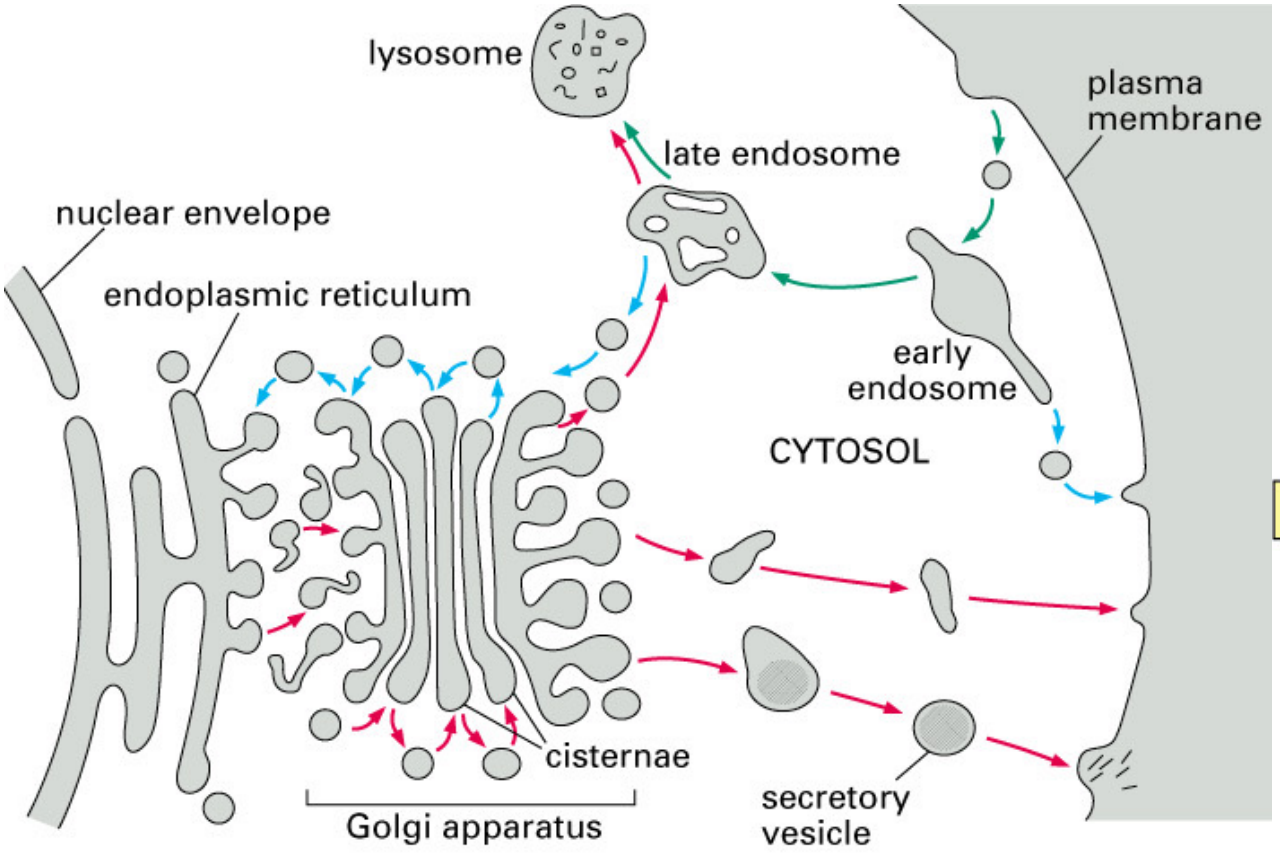


Figure 13–3. Molecular Biology of the Cell, 4th Edition.

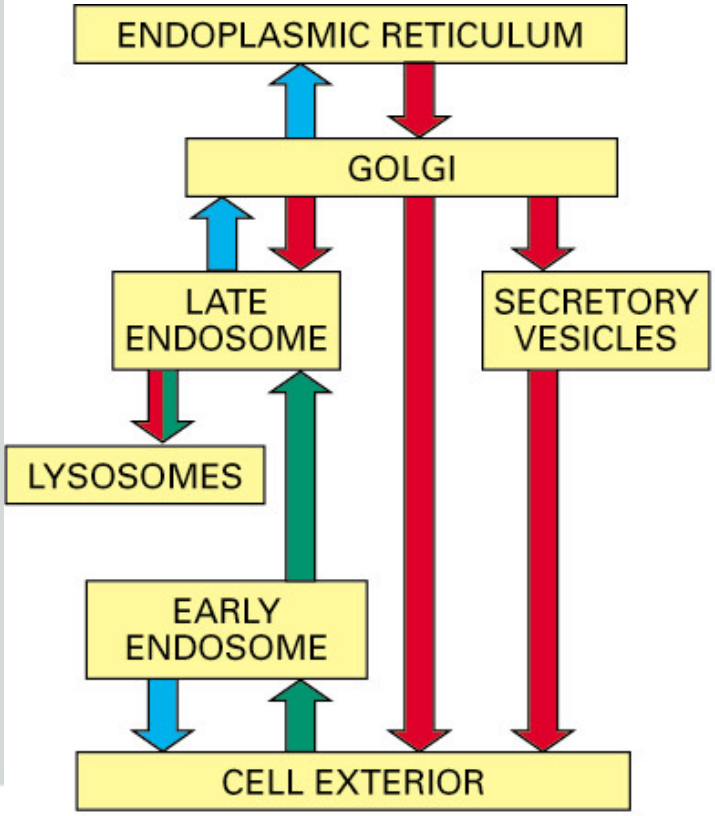
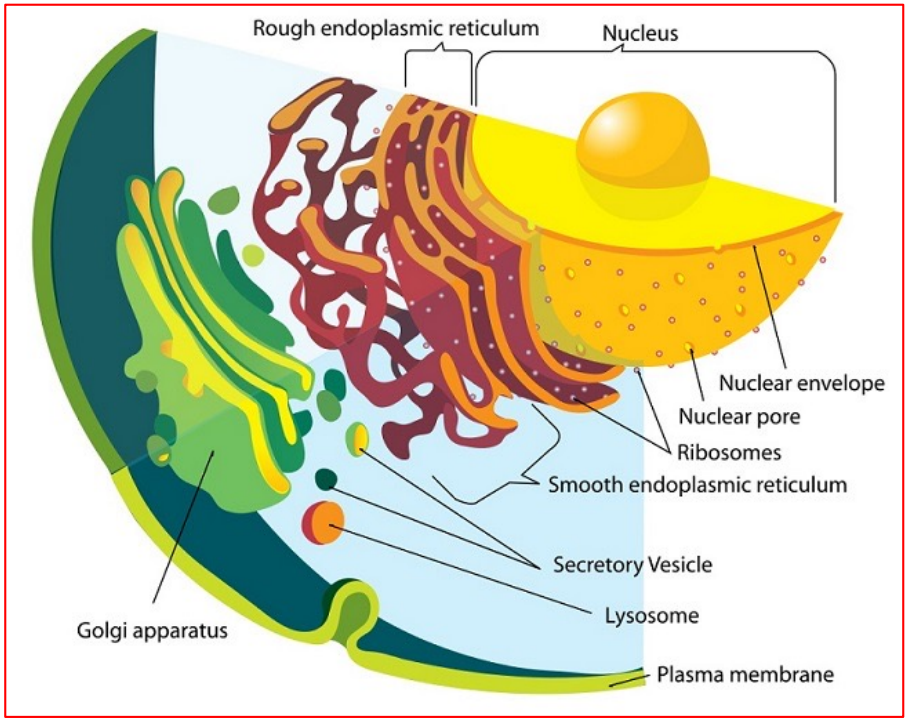
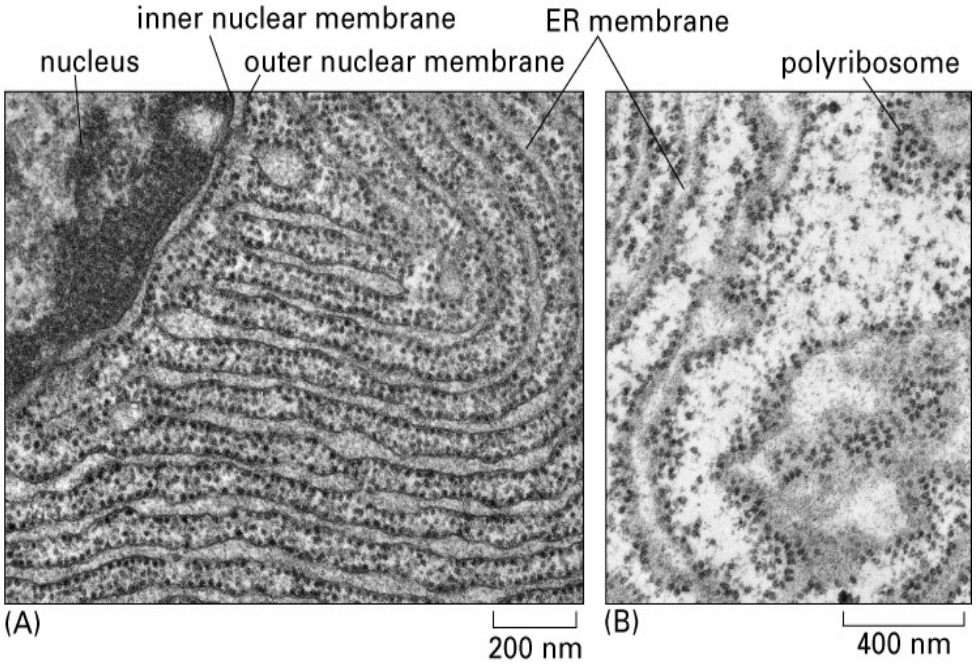


Figure 13–1. Molecular Biology of the Cell, 4th Ed

-They communicate to each other
-Interconnected
-Very dynamic

2 Endoplasmic reticulum (ER)

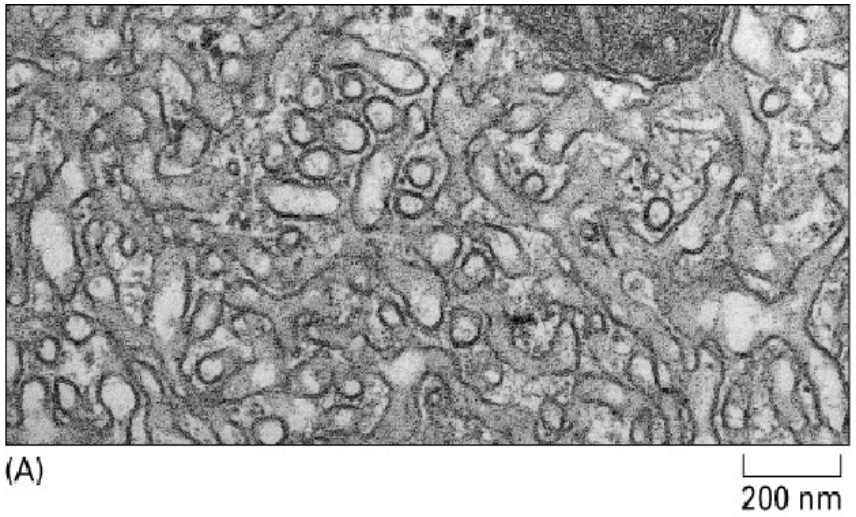




Interconnected

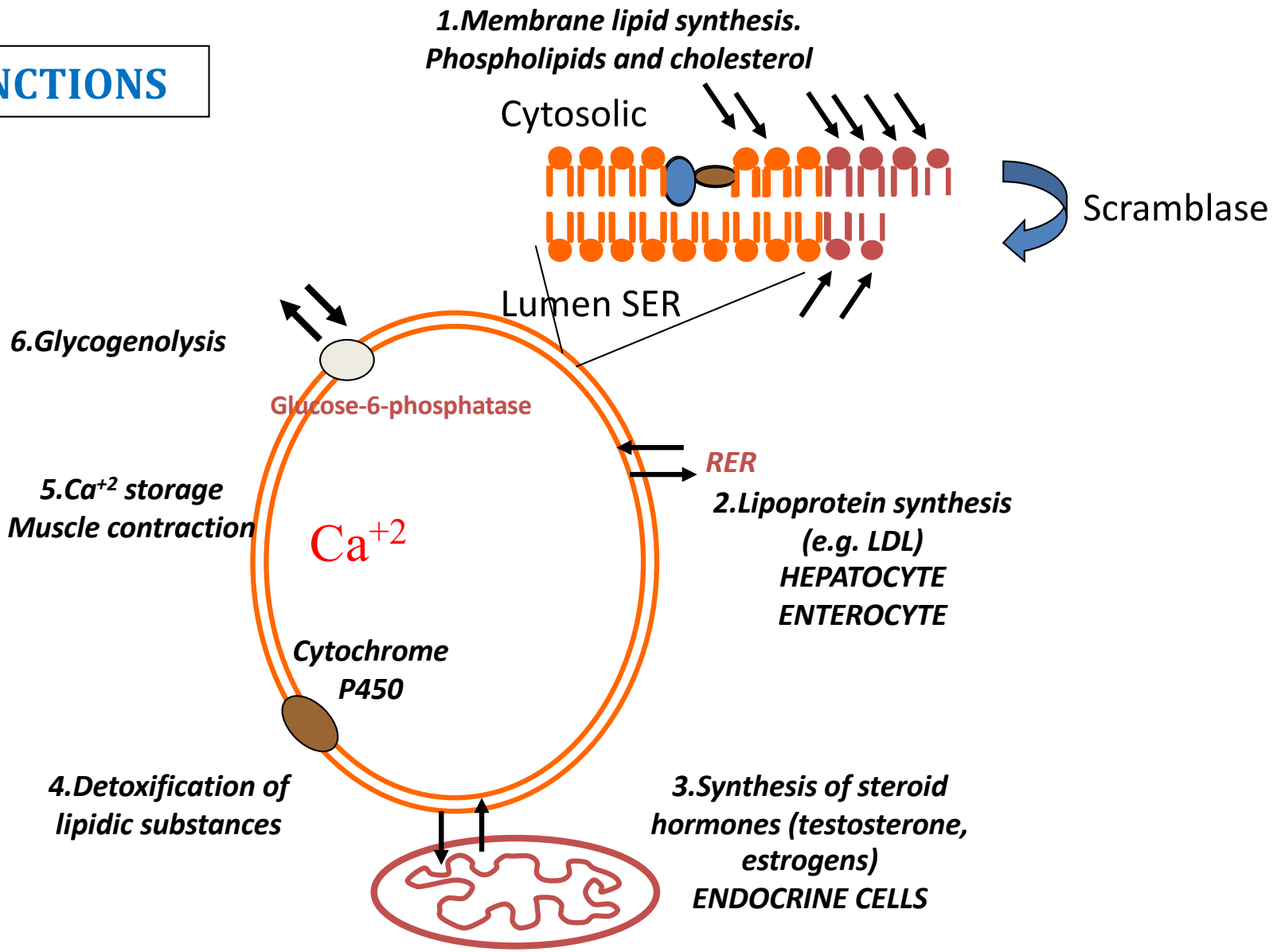
Rough ER:
- Flat parallel sacs
- Ribosomes

Smooth ER:
- Tortuous tubules
- No ribosomes



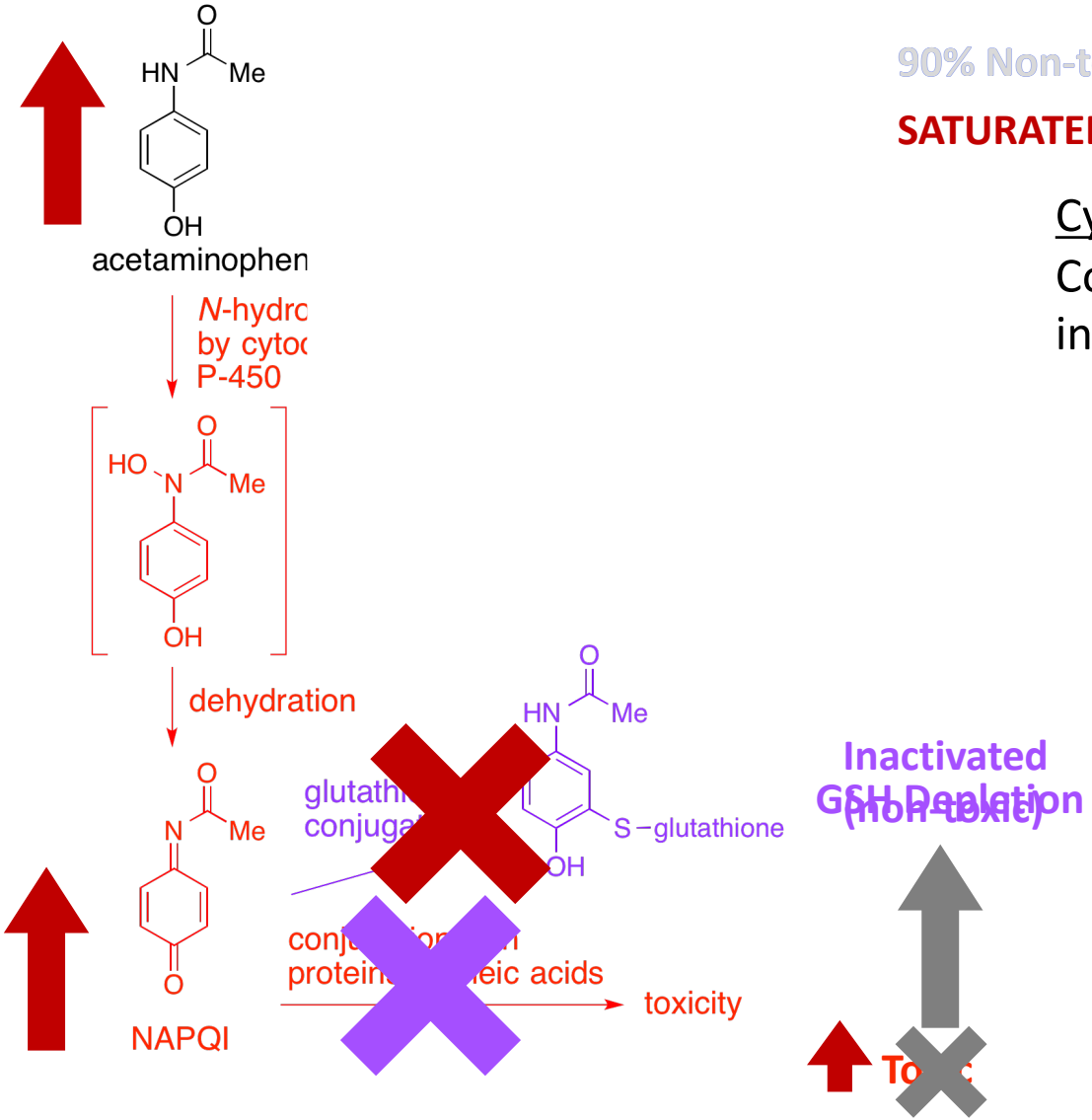
2.1 Smooth endoplasmic reticulum (SER)

FUNCTIONS



2.1 Smooth endoplasmic reticulum (SER)

Detoxification of paracetamol (acetaminophen)



90% Non-toxic

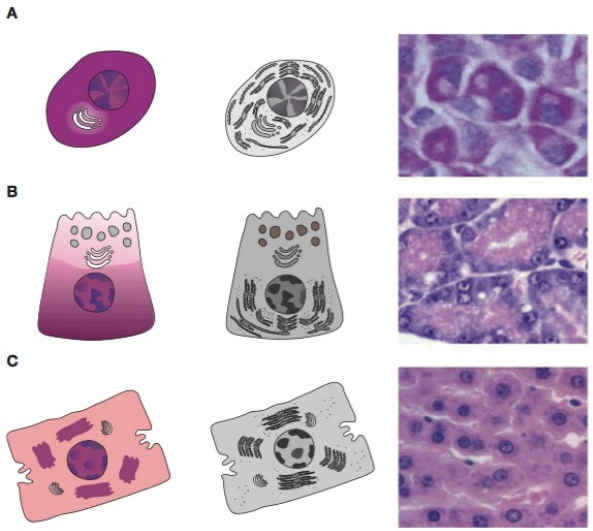
SATURATED PATHWAY

Cytochrome P450
Converts lipidic toxic molecules
in non-toxic water soluble

-Alcohol abuse
-Chronic alcoholism

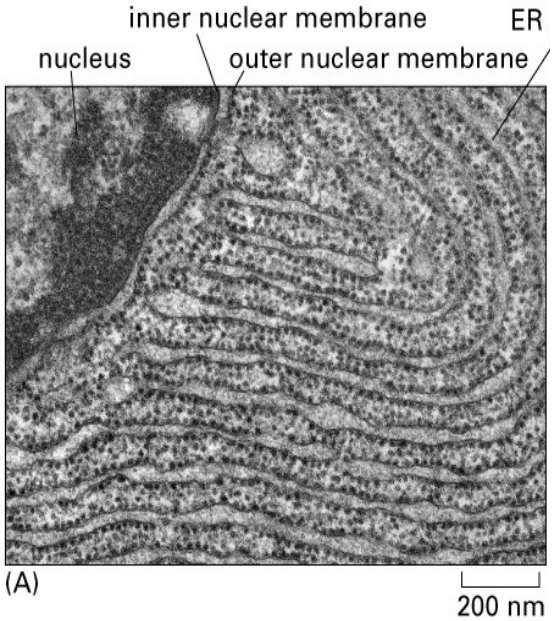
LETHAL DOSE
OF PARACETAMOL

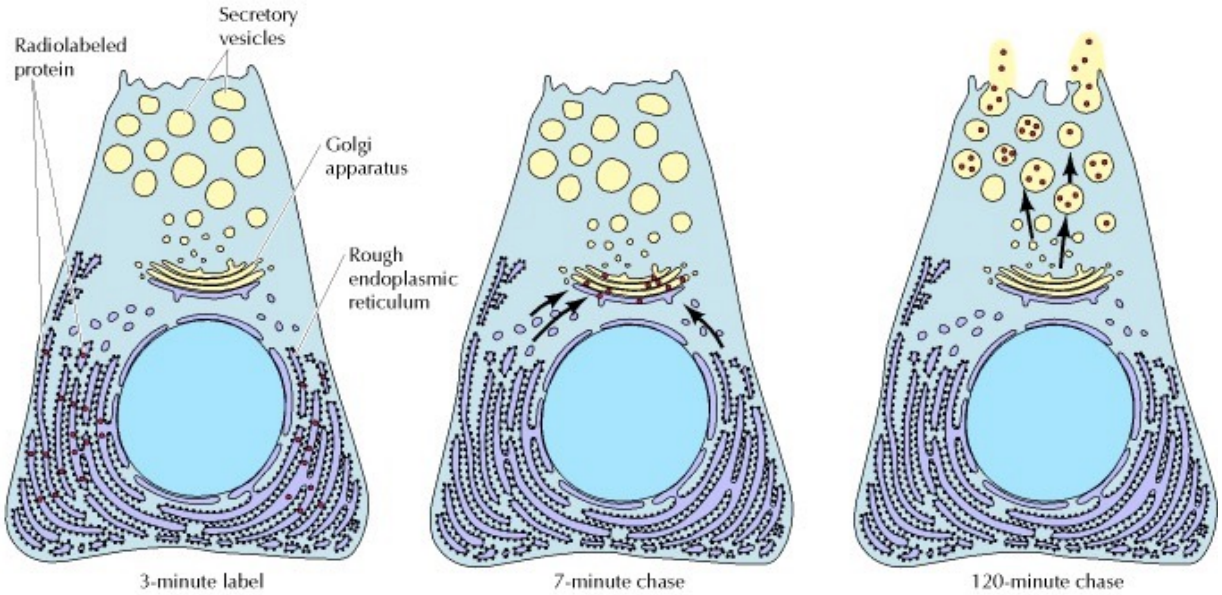
2.2 Rough endoplasmic reticulum (RER)



FUNCTIONS

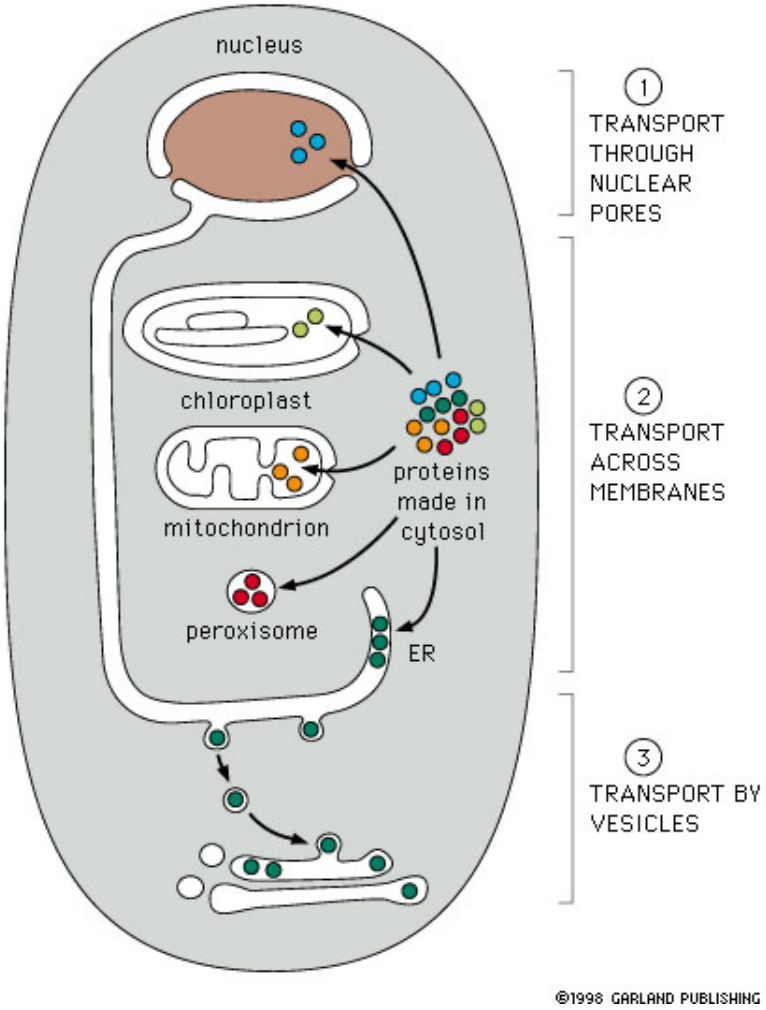
- Participation in protein synthesis and orientation in the membrane.
- protein folding
- Post-translational modifications
  - -N-glycosylation
  - -Disulfide bonds





2.3 Protein trafficking and destination

SYNTHESIS AND DESTINATION OF PROTEINS: **SIGNAL PEPTIDES**



Free ribosomes

- Cytosol (no signal peptide)
- Mitochondria
- Peroxisomes
- Nucleus

With signal peptide

Post-translational importation

With ER signal peptide

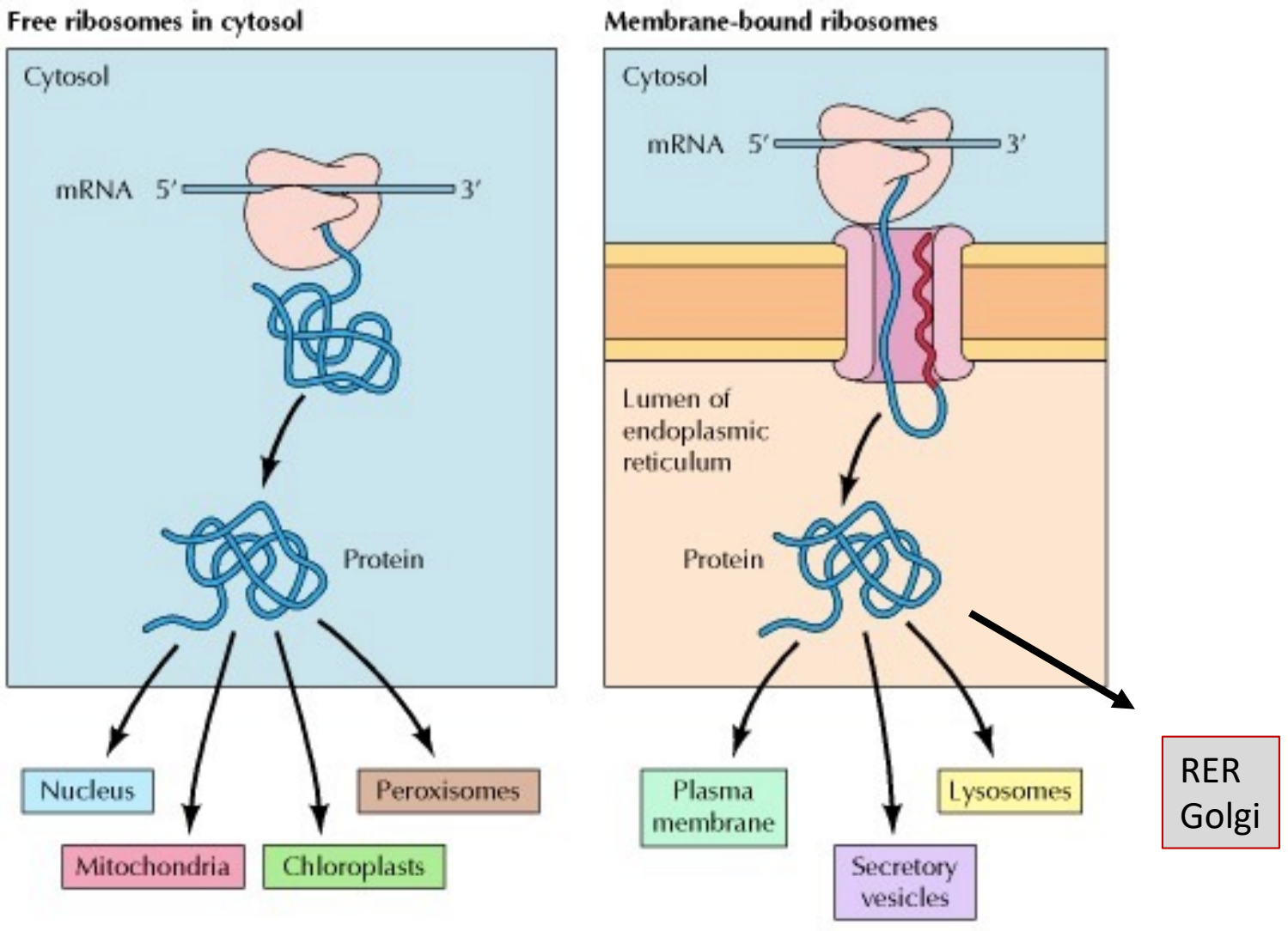
RER

Co-translational importation

- RER / SER
- Golgi
- Lysosomes
- Membrane
- Cell exterior

Specific signals depending on the destination

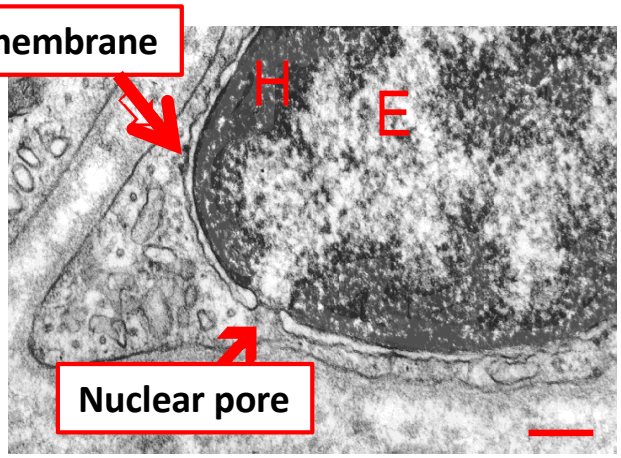
2.3 Protein trafficking and destination

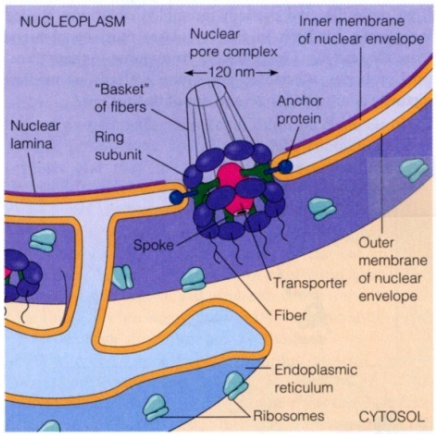


Post-translational importation

Co-translational importation

2.4 Nuclear translocation of cytosolic proteins

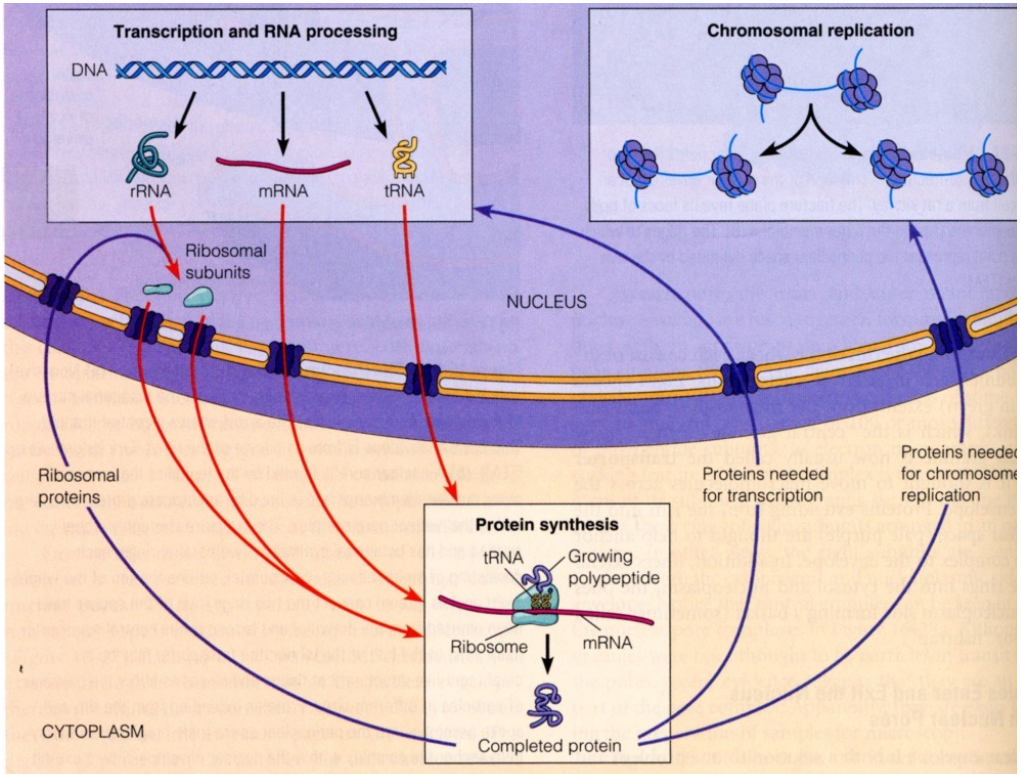




Nuclear localization signal (NLS)

↓

Example: T antigen
(SV-40 viral protein)



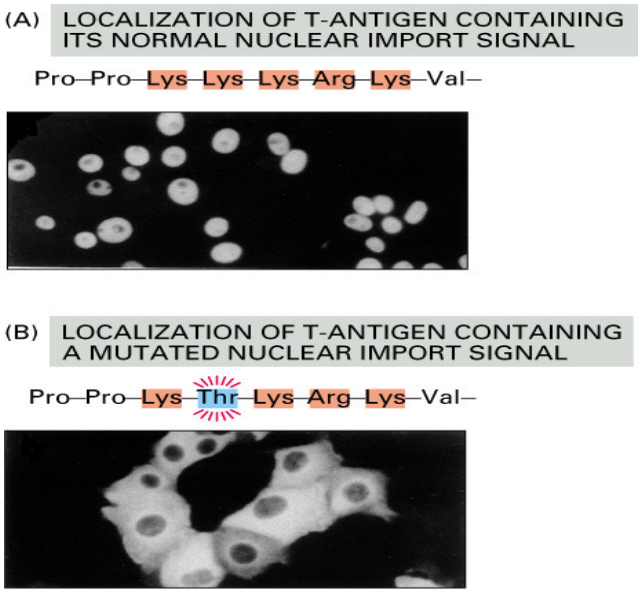
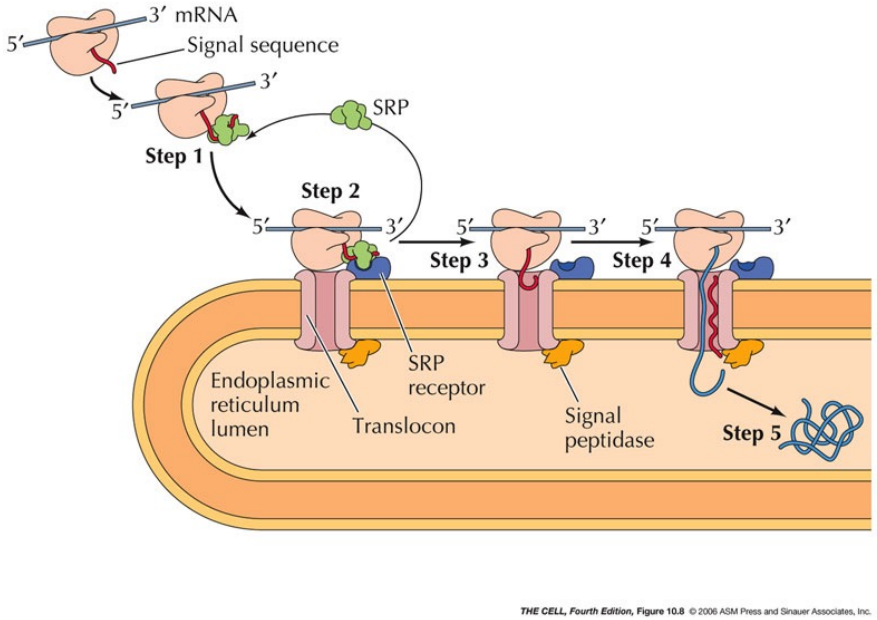


Figure 12–12. Molecular Biology of the Cell, 4th Edition.

2.5 Protein synthesis in the RER

Co-translational importation



THE CELL, Fourth Edition, Figure 10.8 © 2006 ASM Press and Sinauer Associates, Inc.

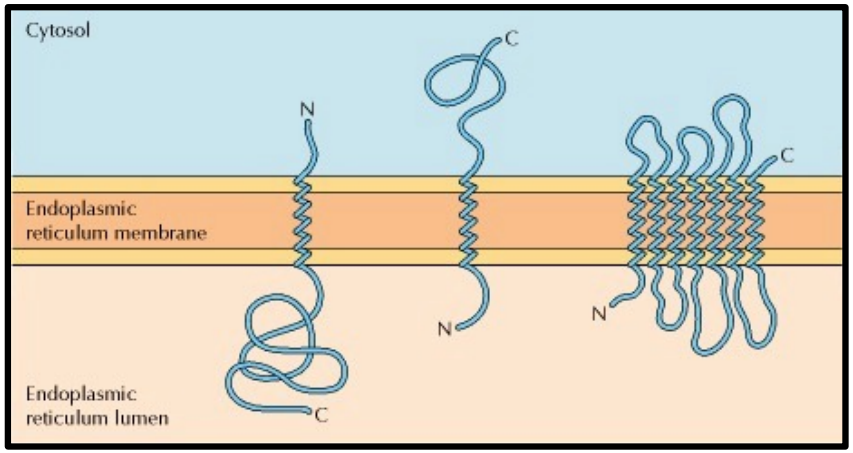
-Signal peptide: 16-30 aminoacid residues

-SRP: Signal recognition particle (N-term). Ribonucleoprotein

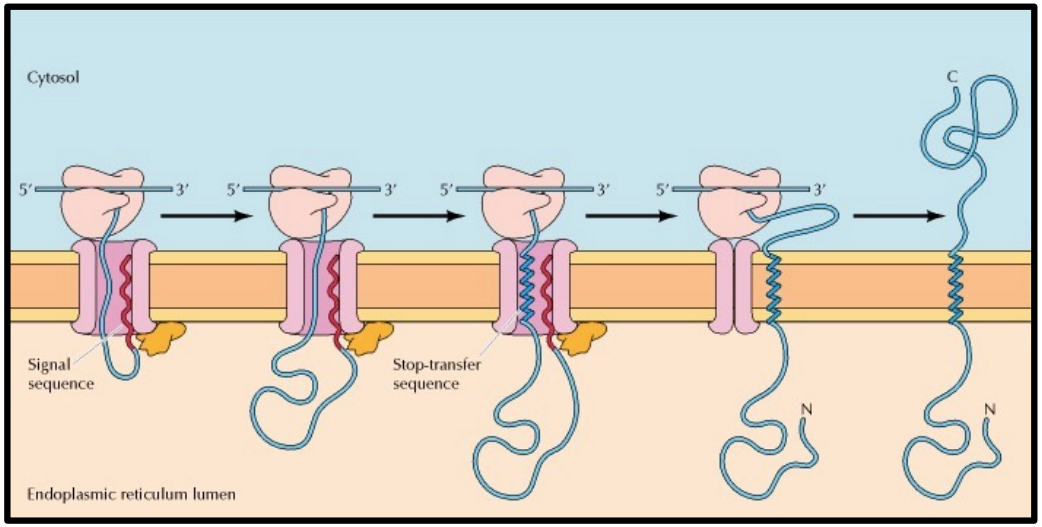
Step 1: Translation starts. Signal peptide recognition by SRP. Translation stops.
Step 2: SRP binds to its receptor. Ribosome/SRP binds to a membrane channel (translocon).
Step 3: Transocon opens and translation resumes.
Step 4: Signal peptide is cleaved (signal peptidase).
Step 5: Translation ends and protein remains in RER lumen.

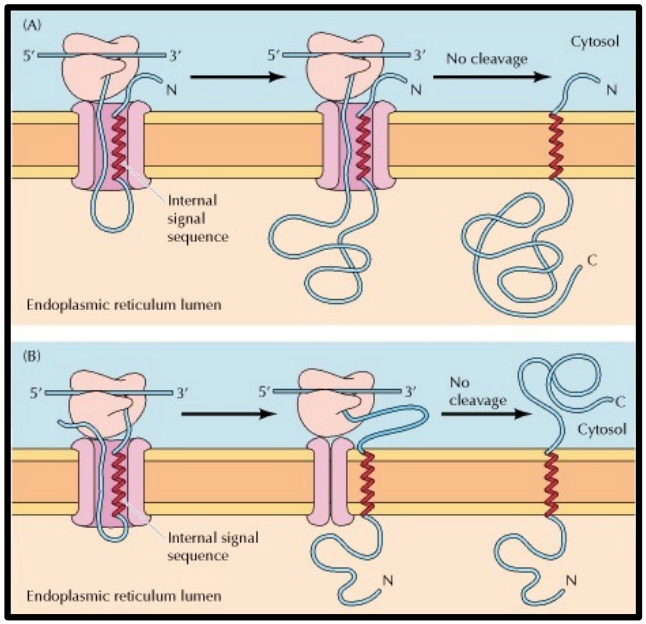
2.5 Protein synthesis in the RER

Co-translational importation



???

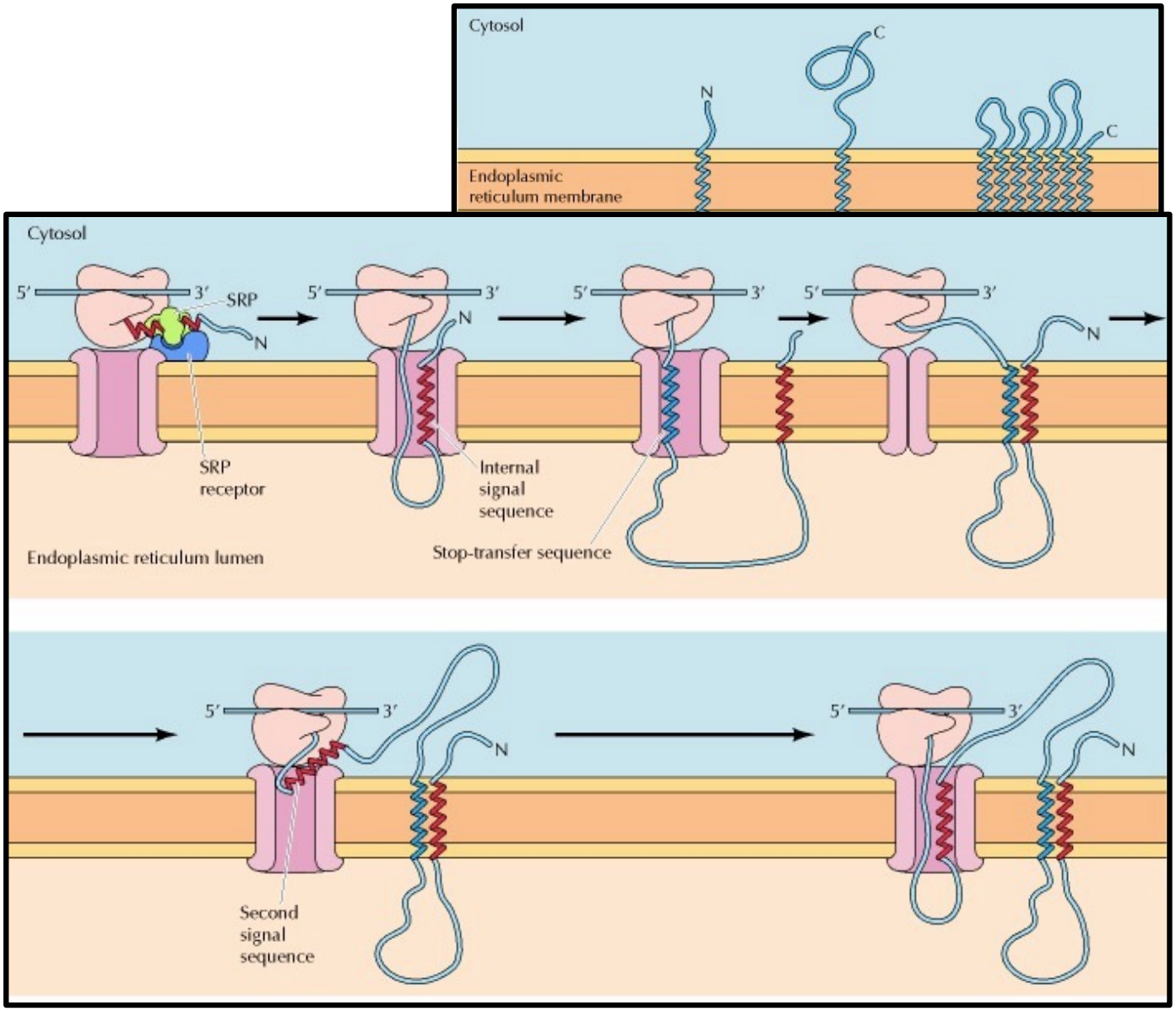




Stop transfer sequence: hydrophobic

2.5 Protein synthesis in the RER

Co-translational importation

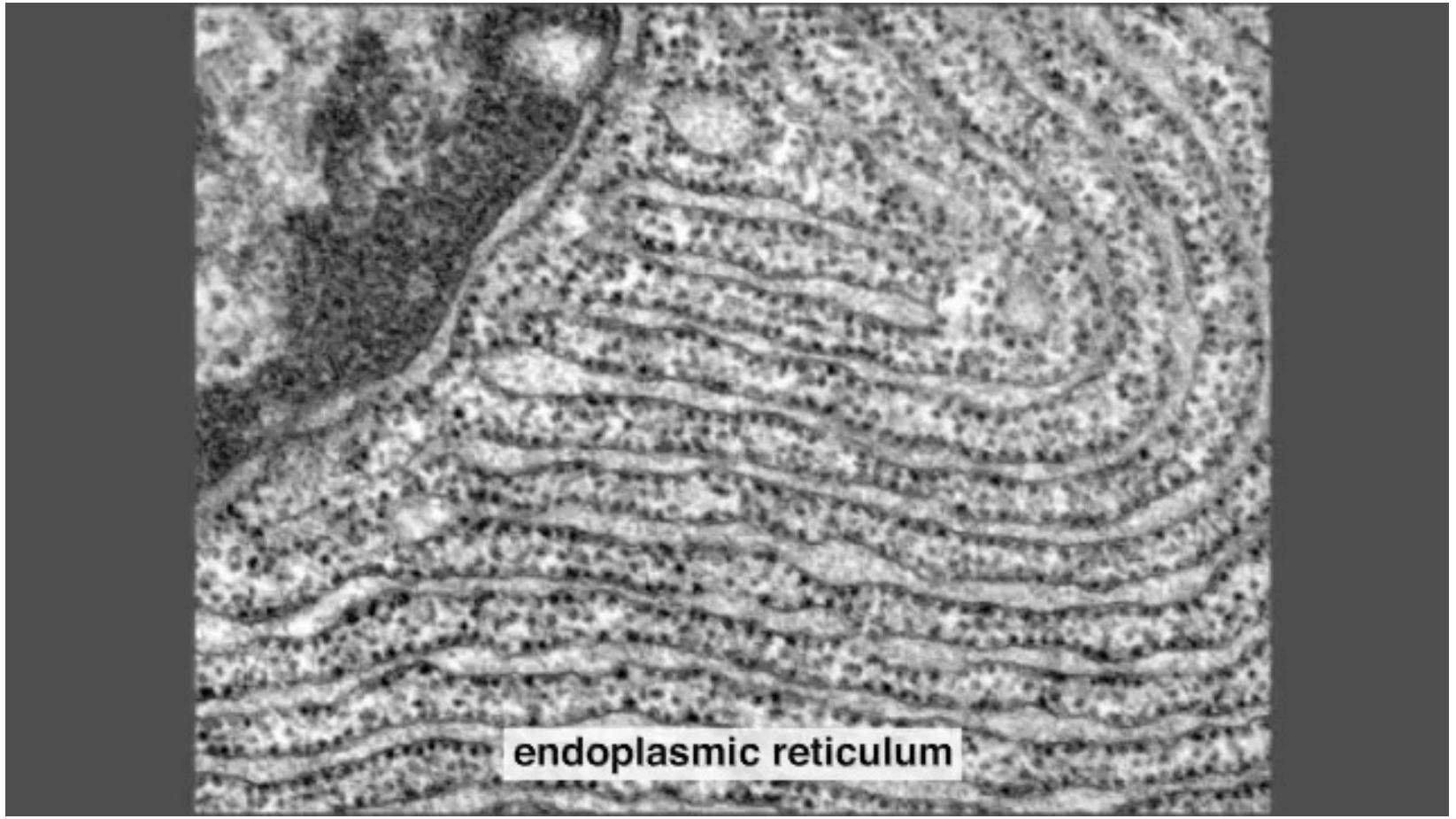


-Several internal signal sequences

-Several stop transfer signals

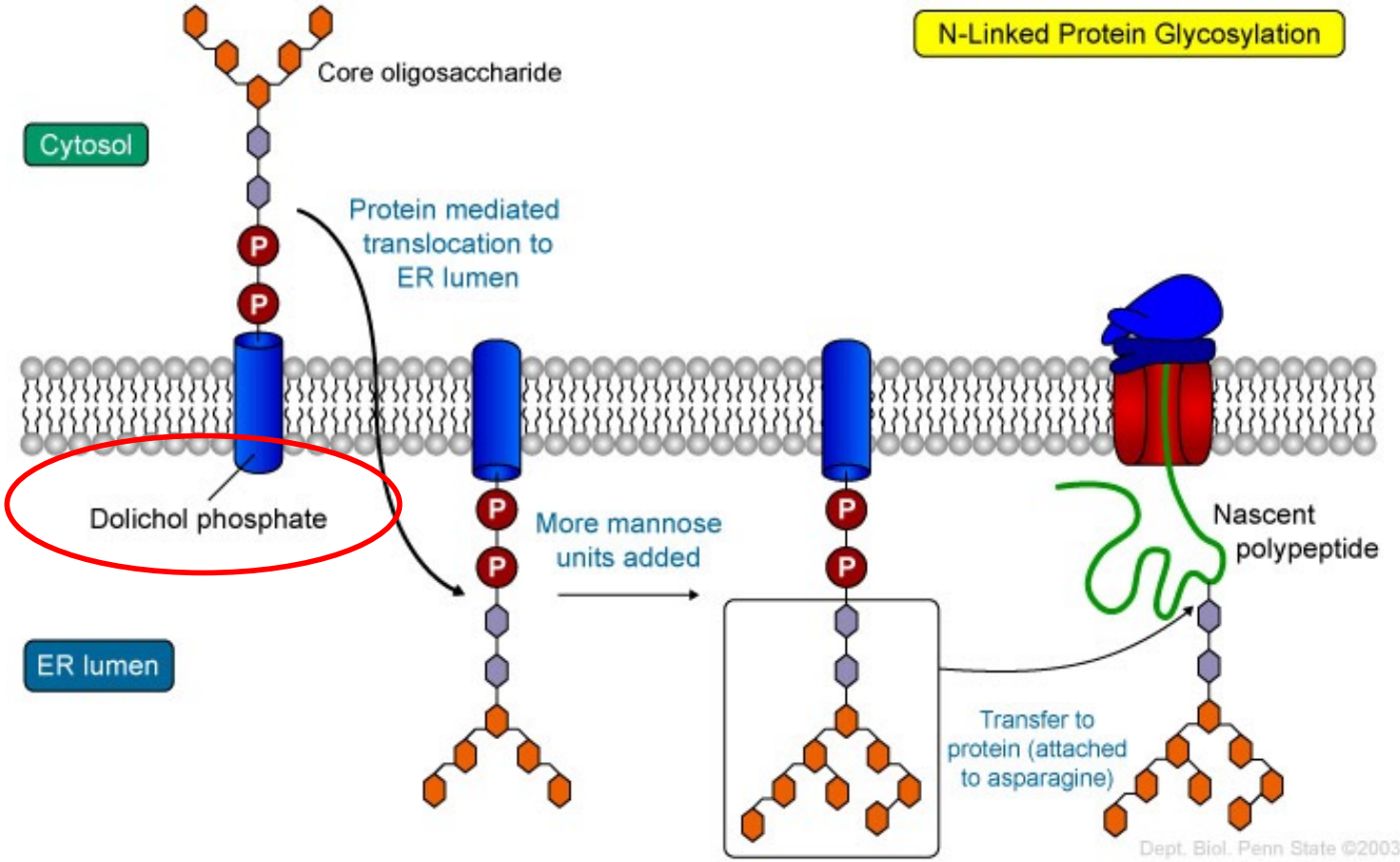
2.5 Protein synthesis in the RER

Co-translational importation



2.5 Protein synthesis in the RER

N-glycosylation in the RER



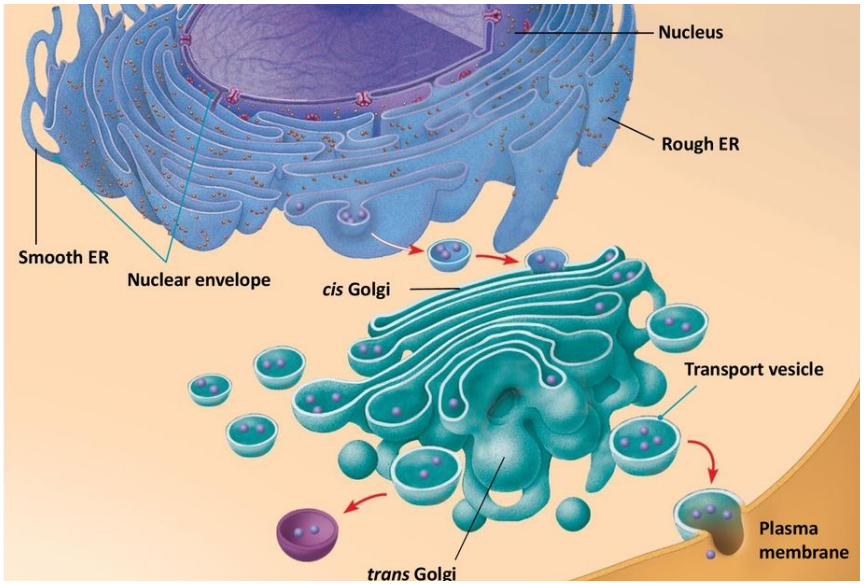
Sequence in the protein needed for the transference:

- Asn-X-Ser
- Asn-X-Thr

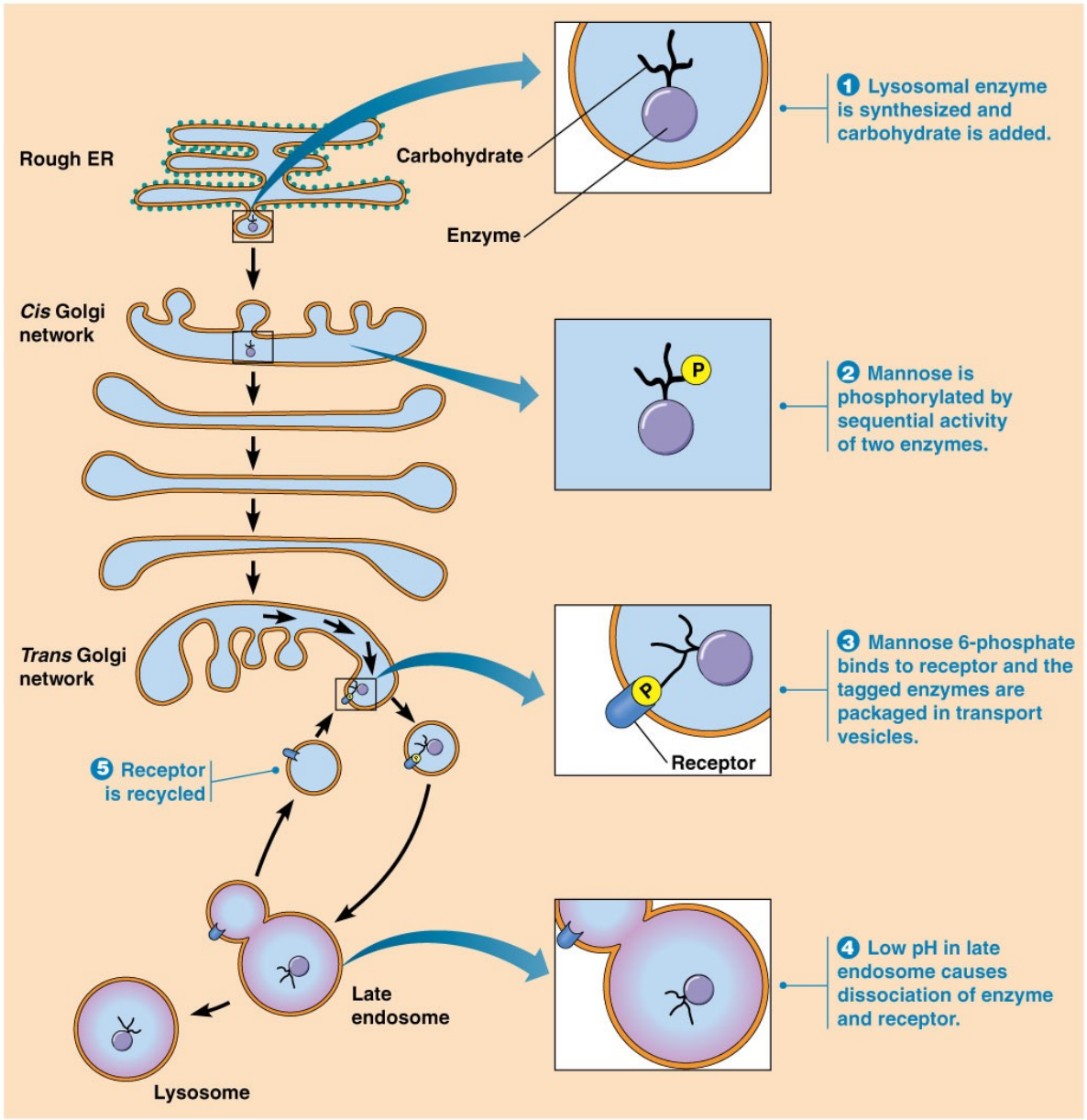
3 The Golgi complex

MAIN FUNCTIONS

- Polarized: cis-golgi, trans-golgi
- Modification of N-glycosylation / O-glycosylation (Ser, Thr).
- Tagging of lysosomal enzymes.
- Formation of secretory vesicles.
- Membrane recycling.
- Distribution of macromolecules



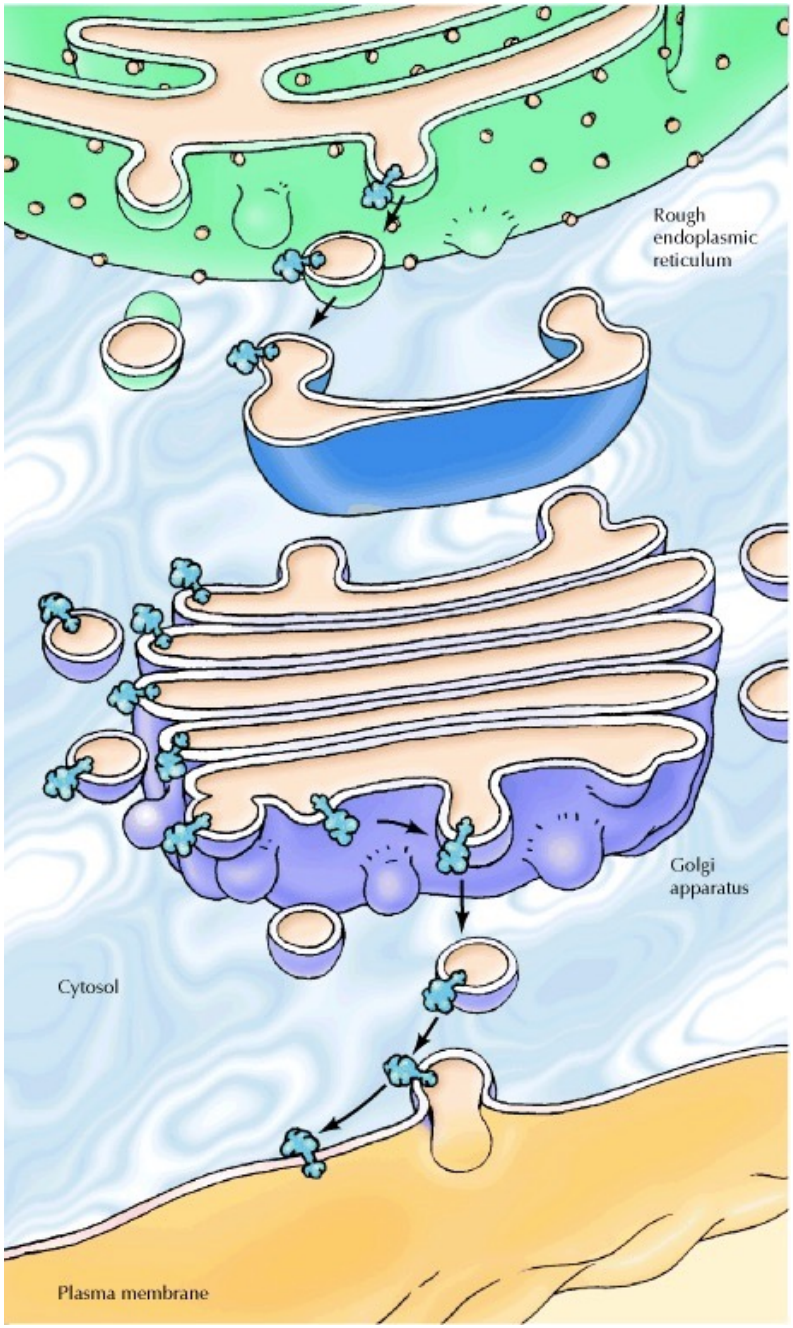
3.1 Example: lysosomal enzymes



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- Synthesis/ N-glycosylation (RER)
- Phosphorylation in mannose (**M6P**)
- Recognition by a receptor
- Travel in vesicles and fusion with Late endosome

4 Protein orientation



Lumens of **endoplasmic reticulum and Golgi** are equivalent to the exterior of cell

Translocated polypeptide chains into RER => exposed to exterior

External portion of receptor

Lumen of ER and Golgi